BIM World

- any project old or new … need?

- 1 Storkholders
- 2 owner
- 3 designer
- 4 consultant
- 5 contractors

History for construction

- create drawing by hand

- 0 need faster
- 1 need copy
- 2 need edit faster
- 3 need accurate information
- 4 need build complex building
- 5 shear information
- 6 work together

CAD

- 1960 Patrick j CAD

- CAD → Computer aided design

- 0 we can faster
- 1 we can copy
- 2 we can edit faster

BIM Tools Timeline

- 1 Auto CAD – 1982 using CAD technology
- 2 Arch CAD – 1982 till now using BIM
- 3 revit – 2002 till now using BIM

What’s BIM?

- Rumors about BIM

- owner – just 3d rending
- arch – just 3d Modling
- const – is coordination and analysis

Design Modeling

- concept
- DESIGN
- ANALYSIS
- SCHEDULING

- construction
- FABRICATION
- construction
- Handover

Operations

- operations
- operations & maintenance
- renovation

BIM Workflow

- 3 Ms to define BIM

- 1 M modeling
- 2 M model
- 3 M management

BIM Process

- BIM is a process

- implementing the process
- define contract like BIM execution
- shear information
- common data environment

Building Information Management

- is the process of designing, constructing
- and operating a building with the use of electronic object-orientated information

BIM Maturity Level

- BIM level 0
- BIM Level 1
- CDE

BIM Level 2

- distinguished by collaborative working and requires information exchange process coordinated between systems and parties

Clash Detection

- Why important?

- identifying, inspecting and reporting interference
- checking work status
- reduce human errors

Benefits

- increasing productivity
- reducing construction cost
- less change order

Clash Detection Software

- navisworks
- tekla bim sight
- solibri
- BIM collab
- BIM Track

BIM Power

- virtual communication
- collaboration with team
- 3D coordination

Phase Planning

- generate accurate program
- show construction sequence
- space requirements
- powerful visualization tool

4D Software

- primavera
- ms project
- navisworks
- synchro 4D